

Recognized by



MSME

MICRO, SMALL & MEDIUM ENTERPRISES

सूक्ष्म, लघु एवं मध्यम उद्यम

OUR STRENGTH • हमारी शक्ति

Ministry of MSME, Govt. of India

startupindia

Internet of Things (IoT) and Machine Learning Training Course

Internet of Things or IOT in short, is the idea of making devices and objects smarter by linking them to the internet. Master the most advanced Electronic, Automation and IoT Cloud technology and build projects using IoT Development Kit. Learn **ESP8266, Arduino, Python, Thingspeak and Blynk** Platforms. This training program introduces you to the amazing world of IOT and its fascinating applications. Using IoT Development kit, you will develop more than 15 Industrial Projects. Build IoT Analytics and Machine Learning Projects.

Syllabus

Module-1: Introduction to Internet of Things

- Introducing to IOT
- IOT Applications
- IOT Network Architecture
- IOT Device Architecture
- IOT Communication Protocols
- IOT Product Development Overview

Module-2: Getting Started with Arduino

- A tour of IoT Development Board Hardware
- Analog and Digital Pins
- Types of Arduino Boards
- Introduction to Arduino programming
- Variables
- IF-Else conditional statements
- Loops: For, While
- Functions Digital Inputs and Digital Outputs
- The serial monitor
- Arrays and strings
- Using Libraries in Arduino
- Arduino data types
- Arduino Commands

Module-3: Sensors, Actuators & Electronics

- Introduction to sensors and types
- Analog Sensors: Temperature, Light Sensor, Potentiometer,
- Digital Sensors: Soil Moisture sensor, DHT11 sensor, Button
- Digital Signals
- Introduction to Actuators
- Interfacing Piezo Buzzer
- Interfacing LED's
- Interfacing RGB LED's

Module-4: Wireless Communication Technologies

- Introduction to wireless technologies
- WiFi
- Bluetooth
- Zigbee
- RFID
- LoRaWAN, SigFox, GSM and NB-IoT
- TCP/IP

Module-5: Cloud Data Monitoring using Arduino

- Concept & Architecture of Cloud
- Role of Cloud Computing in IoT
- Tools, API and Platform for integration of IoT devices with Cloud
- Internet of Things platforms for Arduino
- Posting the sensor data online
- Retrieving your online data
- Monitoring sensor data from a cloud dashboard
- IoT cloud platform and integration with Gateway
- Working with Thingspeak platform

Module-6 : Introduction to MATLAB

- MATLAB Overview
- MATLAB Programming
- MATLAB Basic Operations
- Working with Arrays in MATLAB

Module-7 : Data Analysis using MATLAB

- Statistics using MATLAB
- Outlier Detection
- Z-Score Analysis
- Project: Energy Saving using Anomaly Detection

Module-8: Data Visualization using MATLAB

- Introduction to Data Visualization using MATLAB
- Sensor Data Visualization using MATALB

Module-9: Machine Learning using MATLAB

- Introduction to Machine Learning
- Regression
- Regression in MATLAB
- Predicting Humidity using Regression

Module-10 : IoT using Blynk Mobile Platform

- Setting up Blynk
- Install Blynk Library
- Exploring various control widgets
- Exploring various display widgets
- Notification widgets and virtual pins
- Home Automation Project
- Sensor Data Visualization using Blynk APP

Module-11 : Smart Applets suing IFTTT

- Introduction
- Automate day to day activities through IFTTT
- Posting updates on Facebook
- Automation with IFTTT
- Sending text message notifications
- Google Voice to control Home Appliances

Module-12: IoT Capstone Project

Course Projects, Industry Projects



Smart Home



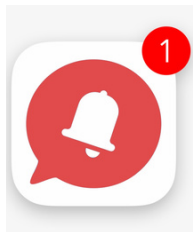
Smart Light Bulb



Voice Controlled Home Automation



Smart Agriculture



IoT based Notifications



Smart Applets



Smart Button



Notification cube

Tools



Training Highlights

- Get Trained by Trainers from IIT Alumni
- Classroom Training
- 10 + projects
- 20+ practical Assignments
- Internship Certification

Why enroll for IoT Certification Course?



By 2020, 50 billion devices will be connected via the Internet of Things (IoT)



Average salary of IoT Developer is \$150k - Glassdoor.com



The global IoT market is expected to grow from \$655.8 billion to \$1.7 trillion in 2020

Course Fee:

RS 5300/- per student

> REGISTER NOW



Website: www.appletoninnovations.com
email: info@appletoninnovations.com /
appletoninnovations@gmail.com